

Department of Mathematics
Motilal Nehru National Institute of Technology Allahabad
Mathematics II (MAN12101)
Course Instructor: Dr. Naren Bag
Assignment 4 (Laplace Transform)

1. Find the Laplace transform of the following functions:

(i) $t^2 \cos t$ (ii) $\sin^3 2t$ (iii) $te^{-3t} \sin t$ (iv) $\frac{\cos \sqrt{t}}{\sqrt{t}}$ (v) $e^{2t} \cos^2 t$ (vi) $\sin t \sin 2t \sin 3t$
(vii) $e^{-3t} \sin 5t \sin 3t$ (viii) $[t]$ (ix) $|t-1| + |t+1|, t \geq 0$

2. Find the Laplace transform of Heaviside function, Dirac delta function and $erf(t)$.

3. Show that $L\left\{\frac{f(t)}{t}\right\} = \int_0^\infty F(s)ds$, where $L\{f(t)\} = F(s)$. Also using Laplace transformation evaluate $\int_0^\infty e^{-t} \left(\frac{\sin^2 t}{t}\right) dt$.

4. Find the Laplace transform of the following periodic functions:

(i) $f(t) = \begin{cases} t/a, & 0 \leq t \leq a \\ (2a-t)/a, & a \leq t \leq 2a. \end{cases}$ (ii) $f(t) = \begin{cases} \cos t, & 0 < t < \pi/2, \\ -\cos t, & \pi/2 < t < 3\pi/2, \\ \cos t, & 3\pi/2 < t < 2\pi. \end{cases}$

5. Find the Laplace transform of the following integrals

(i) $\int_0^t \frac{e^{-t} \sin t}{t} dt$ (ii) $e^{-t} \int_0^t \frac{\sin t}{t} dt$ (iii) $\int_0^t e^{-2t} t \sin^3 t dt$

6. Evaluate the following integrals using Laplace transform

(i) $\int_0^\infty te^{-2t} \sin t dt$ (ii) $\int_0^\infty \frac{\sin mt}{t} dt$ (iii) $\int_0^\infty e^{-t} \left(\frac{\cos at - \cos bt}{t}\right) dt$

6. Solve the following integral equations by using Laplace transform

(i) $y(t) = a \sin t - 2 \int_0^t y(u) \cos(t-u) du$ (ii) $y(t) = a \sin t + 2 \int_0^t y'(u) \sin(t-u) du, y(0) = 0$
(iii) $y'(t) = t + \int_0^t y(t-u) \cos(u) du, y(0) = 4$

7. Find the inverse Laplace transform of the following functions

(i) $\tan^{-1} \frac{2}{s^2}$ (ii) $\cot^{-1} \frac{s}{2}$ (iii) $\log \frac{s^2-1}{s^2}$ (iv) $\log \frac{s^2+1}{s(s+1)}$ (v) $\log \frac{s+1}{(s+2)(s+3)}$ (vi) $\frac{1}{2} \log \left(\frac{s^2+a}{s^2+b}\right)$
(vii) $\frac{5s+3}{(s-1)(s^2+2s+5)}$ (viii) $\frac{s}{s^4+s^2+1}$ (ix) $\frac{1}{s^3(s^2+a^2)}$ (x) $\frac{1}{(s^2+a^2)^{3/2}}$ (xi) $\frac{1}{s^2-4s+8}$

8. Apply convolution theorem to evaluate

(i) $L^{-1} \left\{ \frac{s^2}{(s^2+a^2)(s^2+b^2)} \right\}$ (ii) $L^{-1} \left\{ \frac{s}{(s^2+1)(s^2+4)(s^2+9)} \right\}$ (iii) $L^{-1} \left\{ \frac{1}{(s^2+4s+13)^2} \right\}$ (iv) $L^{-1} \left\{ \frac{s}{(s^2+a^2)^3} \right\}$

9. Find solution of the following initial value problems by using Laplace transform

(i) $x'' + 2x' + 5x = e^{-t} \sin t; x(0) = 0, x'(0) = 1.$ (ii) $tx'' + x' + tx = 0; x(0) = 2, x'(0) = 0.$
(iii) $x''' - 3x'' + 3x' - x = t^2 e^{2t}; x(0) = 1, x'(0) = 0, x''(0) = -2.$
(iv) $tx'' + 2x' + tx = \cos t; x(0) = 1.$ (v) $tx'' + (1-2t)x' - 2x = 0; x(0) = 1, x'(0) = 2.$

10. Solve the following initial-boundary value problems.

(i) $u_x + xu_t = 0, u(x, 0) = 1, u(0, t) = t.$ (ii) $u_x + 6xu_t = 12x, u(x, 0) = 2, u(0, t) = 2.$
(iii) $u_{tt} = u_{xx}, 0 < x < l, t > 0; u(x, 0) = 0, u_t(x, 0) = \sin(\pi x/l), 0 < x < l; u(0, t) = 0,$

$$u(l, t) = 0, \quad t > 0.$$

$$\text{(iv)} \quad u_{tt} = u_{xx}, \quad x > 0, \quad t > 0; \quad u(x, 0) = e^{-x}, \quad u_t(x, 0) = 0, \quad u(0, t) = 0, \quad u(x, t) \text{ is bounded as } x \rightarrow \infty.$$